

Development Kickoff : 18/09/2025

In the beginning, we brainstormed concepts for potential project ideas, drawing inspiration from various real-life application examples presented in today's lecture.

The first idea came from Fatama: bringing fantasy books to life by projecting 3D models of various animals from within a given book. In addition, we plan to add a button for sound effects corresponding to these animals.

The main consumers for this software would primarily be children, along with their parents. Imagine a scenario where a parent is reading a book to their child: using our software, the child would be able to see a 3D version of the animal and hear the related sound effect.

Our second idea came from Kim. He wanted to address the financial challenges of owning a physical location to sell goods, while also providing the benefit of showing the dimensions of a given product in real life.

In this scenario, the store owner would rent or purchase wall space and display images of products. Consumers could then scan the images with their phones to visualize the products in 3D and, if they wished, purchase them directly.

After some discussion, we decided to go with the first option and began development immediately. However, we also noted that the two projects are similar enough that we could pivot in the future if we decided to change direction.

We started by familiarizing ourselves with Unity and learning how Gitflow works, which was relatively simple to grasp with our teacher's guidance. We set up a system where each team member had their own scene or workspace to avoid conflicts with one another's work, with the idea of later copying useful elements into the main scene that would serve as our final product.

Moving forward, we also decided that each time we meet, we will work together on one scene projected on the big screen. This approach allows us to develop faster and maintain a clear understanding of our overall progress. Today, we made significant progress and managed to get a working concept operational, with the help of YouTube tutorials and ChatGPT.

To get our initial prototype running, we used the following setup in Unity:

- Added pictures of animals into the **Assets** folder to be used as markers.
- Changed the **Texture Type** of each picture to **Sprite (2D and Single)** so they could be used in the Reference Image Library.
- Created a **Reference Image Library** in Unity to store and manage these pictures for AR tracking.
- Linked the library to the **AR Tracked Image Manager** component, enabling the app to detect the pictures in real time.
- Assigned **3D model prefabs** to appear whenever a corresponding picture was detected.